Optimizing Portering Logistics at HHH

An AI-Driven Porter Dispatch System

Team Members

CHAN Hok Nam
CHOI Cheuk Man
KIM Youseung

Supervisor

Prof. Xiangtong QI

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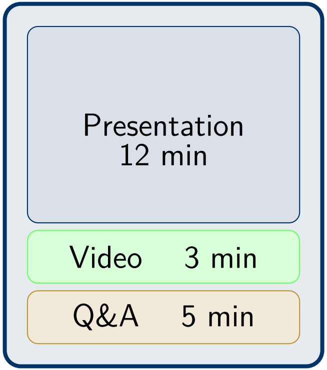
Department of Industrial Engineering & Decision Analytics, HKUST

Presentation Outline

1. **The Problem** — Why does this matter?
2. **Our Solution** — System architecture
3. **How accurate is the AI parsing?**
4. **Does the optimizer actually help?**
5. **Live demo highlights**
6. **Conclusion & Future work**



The Problem: Porter Dispatch at HHH

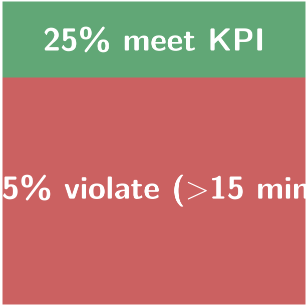
What porters do:

- Transport patients, specimens, equipment between departments
- Operate 24/7 across entire hospital
- Dispatched manually by radio

HHH's stated KPI:

$$\boxed{\textbf{Every task} \leq \textbf{15 minutes}}$$

75% of tasks FAIL the KPI

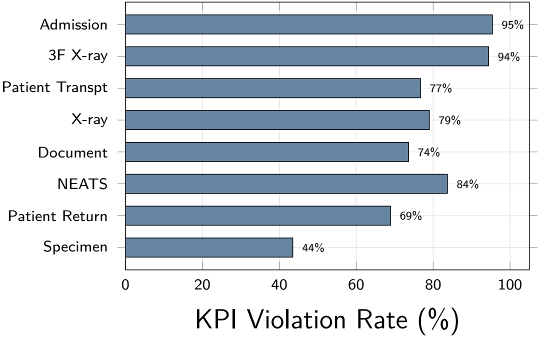


2,843 historical tasks, 2024

Mean duration: 49.7 min

Root Causes of Poor KPI Performance

By service type (violation rate):



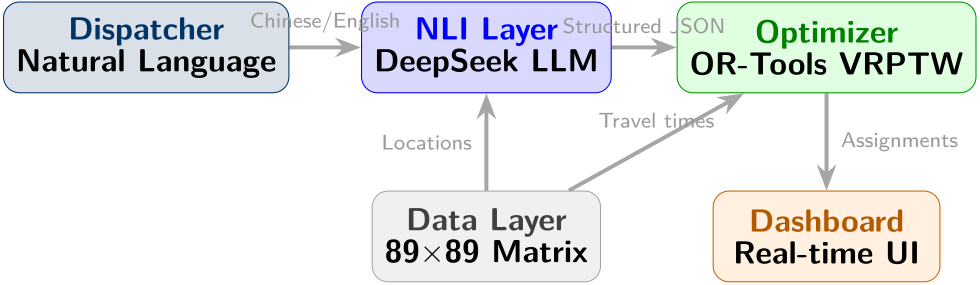
Key insight

High violation rates for admission (95.4%) and 3F X-ray (94.4%) reflect **long fixed travel distances** — not dispatcher error.

Three dispatch problems:

- ✗ Manual, radio-based assignment
- ✗ No global fleet optimisation
- ✗ No structured data capture

Our Solution: Hybrid AI Dispatch System



Flask REST API · Vanilla JS · Python 3

What the System Does — End to End

Step 1 — Dispatcher types:

"Super urgent patient from A&D to SDU"

Step 2 — LLM parses:

```
{from: "A&D", to: "SDU",
 service: "admission",
 priority: "Super-Urgent"}
```

Step 3 — Solver assigns:

Porter P-003 (nearest, 1.2 min away)
Estimated total: **8.4 min**

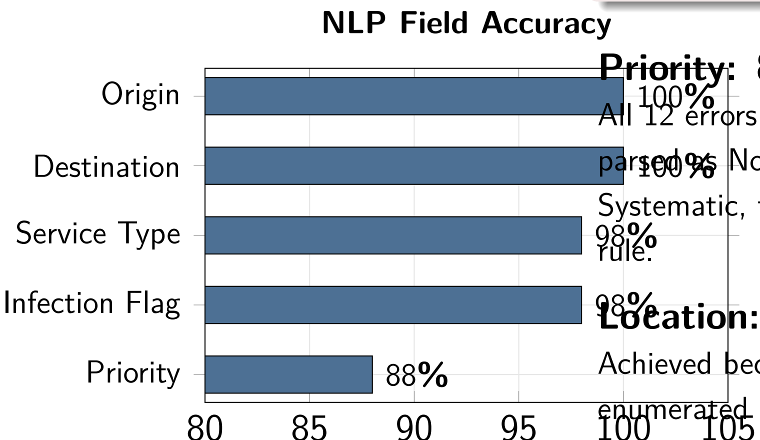
Features at a glance:

- ✓ Chinese & English NL input
- ✓ Multi-task from one request
- ✓ Intermediate stops / waypoints
- ✓ Infection control flags (VRE, NEATS)
- ✓ Scheduled future tasks
- ✓ Rolling-horizon re-optimisation
- ✓ LLM "Why this porter?" explanation
- ✓ RAG policy advisor

How Accurate Is the LLM Parsing? [Tested on 100 samples]

84% all-5-fields correct

Zero LLM parse failures across 100 samples.



Priority: 88%

All 12 errors: "Immediate" (*ji shi*) parsed as Normal instead of Urgent. Systematic, fixable with one prompt rule.

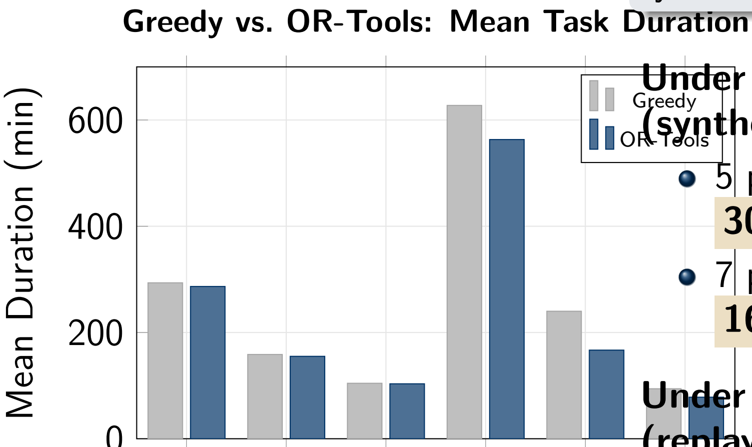
Location: 100%

Achieved because all 89 locations are enumerated in the system prompt.

Does the Optimizer Actually Help?

Key finding

OR-Tools helps most when the system is **not saturated**.

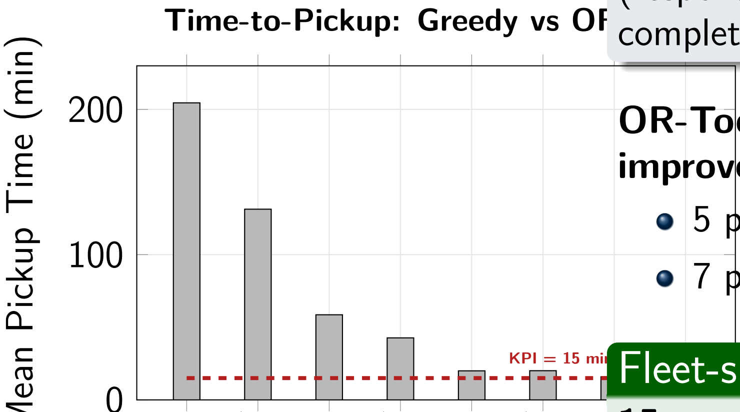


Under light load (synthetic):

- 5 porters: **30.5% improvement**
- 7 porters: **16.9% improvement**

Under real HHH load (replay):

What Does the 15-min KPI Actually Mean?



KPI definition

15 min = porter **arrives at pickup** (response time), not task completion.

OR-Tools pickup improvement:

- 5 porters: **35.8% faster**
- 7 porters: **27.2% faster**

Fleet-sizing finding

System Demo Highlights

Real-time Dashboard

Porter Status (10)

- AVAILABLE
- BUSY
- AVAILABLE
- BUSY
- AVAILABLE

Dispatch Task

"Super urgent patient from A&D to SDU"

Dispatch Task

P-003 assigned | 8.4 min
Why? Nearest to A&D (1.2 min)

Scheduled Tasks

No scheduled tasks.

Policy Advisor (RAG)

KPI Risk Assessment

HIGH RISK — 3F X-ray task, 2 porters busy

Historical mean for this route: 156.8 min

KPI violation rate: 94.4%

Suggestion: pre-assign or flag for supervisor

Ask Advisor

"Which service type needs more porters?"

Based on 2,843 records: 3F X-ray has mean 156.8 min and 94.4% KPI violation. Consider dedicated porter or batching.

Summary of Contributions

1. Validated NLI (LLM layer)

- 84% all-field accuracy on 100 real samples
- 100% location extraction
- Handles Chinese, multi-task, constraints

2. OR-Tools VRPTW Solver

- Pickup time: up to **35.8%** faster than greedy
- Total duration: up to **30.5%**

Deliverables:

- ✓ Deployed web app (Flask)
- ✓ 89×89 travel matrix
- ✓ Discrete-event simulator
- ✓ NLP accuracy test harness
- ✓ Policy advisor (RAG)
- ✓ Simulation benchmarks

Thank You

Questions Welcome

CHAN Hok Nam CHOI Cheuk Man KIM Youseung

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BACKUP SLIDES

for Q&A reference

Backup: OR-Tools Formulation Details

Node graph:

- m porter start depots
- n pickup nodes (task origins)
- n delivery nodes (task destinations)
- 1 dummy end depot

Objective:

$$\min \sum_{(i,j)} c_{ij} + \lambda \sum_{k} \max(0, \mathrm{done}_k - 15)$$

$\lambda = 1000$ centi-min per minute over KPI

Constraints:

- Pickup before delivery (same porter)
- Capacity: 1 task at a time per porter
- Drop penalty $2 \times 10^7 \gg$ any KPI penalty
- 1-second solve limit

Search strategy: PARALLEL_CHEAPEST_INSERTION
$\Rightarrow$ GUIDED_LOCAL_SEARCH

Fallback: Greedy (nearest

Backup: Why Not Train a Custom NLP Model?

Option A: Trained classifier

- ✗ Requires labelled training data
- ✗ 84,000 records need annotation
- ✗ Domain shift if terminology changes
- ✓ Higher ceiling accuracy

Option B: LLM with prompt (chosen)

- ✓ No annotation required
- ✓ Handles unseen phrasings
- ✓ Easy to update (change

Practical verdict

For a hospital prototype without labelled data, the LLM approach is the right engineering choice. 84% all-field accuracy is sufficient for dispatch with human oversight.

A trained model would be the natural next step if HHH commits to deployment and provides

Backup: Pickup-Time KPI Analysis

Porters	Strategy	Mean Pickup (min)	Pickup Violation (%)
5	Greedy	204.5	96.0%
5	OR-Tools	**131.3**	**83.0%**
7	Greedy	58.6	94.0%
7	OR-Tools	**42.7**	**78.0%**
10	Greedy	20.0	59.0%
10	OR-Tools	20.1	60.0%
15	Either	**15.9**	**48.0%**

Synthetic mode · 100 tasks · 8-min mean inter-arrival (historically realistic)

56.6% of all routes have direct travel + service time >15 min — KPI blocks are geographic for these tasks

Backup: Full Simulation Data

Mode	Strategy	Porters	Mean (min)	KPI Viol. (%)
Replay	Greedy	3	293.5	99.0%
Replay	OR-Tools	3	286.6	99.0%
Replay	Greedy	5	158.6	97.0%
Replay	OR-Tools	5	155.1	96.8%
Replay	Greedy	7	104.4	96.0%
Replay	OR-Tools	7	103.4	95.8%
Synthetic	Greedy	3	627.4	100.0%
Synthetic	OR-Tools	3	563.2	100.0%
Synthetic	Greedy	5	240.0	100.0%
Synthetic	OR-Tools	5	166.8	100.0%
Synthetic	Greedy	7	94.1	100.0%
Synthetic	OR-Tools	7	78.2	100.0%